Tokens and Context

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

WORKBOOK · 1 OF 26

Recall and choose your starting point

> Record your starting explanation for Tokens and Context before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

A Distribution of Possibilities

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Tokens and Context trace and record the first step you can explain without help.

Use W2: Choose complete support for Context limits and **evidence selection**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Tokens and Context explanation against the supplied facts before discussing it.

Use the worked case for Context limits and **evidence selection**. Proposed explanation: "**Token** count equals word count". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Budget new evidence independently

Total 120, instructions 25, question 15, **output** reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18.

Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Tokens and Context target to a trace and a justified conclusion.

A toy system allows 12 **token**s total for **input** and **reserved output**. Instructions use 3, a source uses 7 and the planned answer needs 4. Compute the excess and propose a revision that preserves **evidence**.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Tokens and Context case and show the deciding evidence.

A toy window holds eight **token**s; prompt uses six and quotation uses three.

> Does it fit? What should be preserved?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 26

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A system can use only the last two messages. Message 1 says use centimetres; messages 2 and 3 discuss plants. What instruction may be lost, and how can the next request repair this?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 3 Test and reflect

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	

Test record	My evidence
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 26

Project checkpoint

> Justify what Tokens and Context adds to Verified Study Buddy and keep the evidence.

Keep the learner's question, approved source excerpt, and **output** rules inside the active context.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A toy system allows 12 **token**s total for **input** and **reserved output**. Instructions use 3, a source uses 7 and the planned answer needs 4. Compute the excess and propose a revision that preserves **evidence**.

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Compute available evidence

> Compute available evidence. Apply the supplied Tokens and Context facts and keep your reasoning.

Under total 100, instructions 20, question 10 and **output** reserve 20, calculate **evidence** capacity.

My reasoning and evidence

W2 Choose complete support

> Choose complete support. Apply the supplied Tokens and Context facts and keep your reasoning.

Compare A+B,A+C,C+B for fit and support of current approval and capacity.

My reasoning and evidence

W3 Preserve a qualifier

> Preserve a qualifier. Apply the supplied Tokens and Context facts and keep your reasoning.

A passage says room B normally holds 12, except during maintenance when the approved limit is 8. Why is retaining only the first clause inadequate?

My reasoning and evidence

W4 Budget new evidence independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Budget new evidence independently. Apply the supplied Tokens and Context facts and keep your reasoning.

Total 120, instructions 25, question 15, **output** reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18.

Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

My reasoning and evidence

WORKBOOK · 16 OF 26

W5 Audit inclusion

Audit inclusion. Apply the supplied Tokens and Context facts and keep your reasoning.

A team says every claim is verified because all chosen documents fit in context. What further check is needed?

My reasoning and evidence

WORKBOOK · 17 OF 26

Attention as weighted information selection — W1 Identify the roles

Use the supplied Tokens and Context task to record your reasoning, evidence and next step.

Explain **query**, key, **value** and weight without calling the largest weight a truth score.

My reasoning and evidence

WORKBOOK · 18 OF 26

Attention as weighted information selection — W2 Compute the mixture

Use the supplied Tokens and Context task to record your reasoning, evidence and next step.

Calculate the **output** from the three supplied **value**s and weights.

My reasoning and evidence

WORKBOOK · 19 OF 26

Attention as weighted information selection — W3 Mask and renormalise

Use the supplied Tokens and Context task to record your reasoning, evidence and next step.

Exclude V3 and calculate permitted weights and **output**. Explain the empty-**mask** case.

My reasoning and evidence

WORKBOOK · 20 OF 26

Attention as weighted information selection — W4 Combine a new case independently

> Use the supplied Tokens and Context task to record your reasoning, evidence and next step.

Values A=[4,1],B=[0,5],C=[2,3]; weights 0.25,0.5,0.25. Compute the full **output**. **Mask** B, renormalise the remaining weights and compute the new **output**.

My reasoning and evidence

WORKBOOK · 21 OF 26

Attention as weighted information selection — W5 Audit an explanation

> Use the supplied Tokens and Context task to record your reasoning, evidence and next step.

A student claims the highest attention weight explains the entire model and authenticates a retrieved instruction. What additional reasoning is needed?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

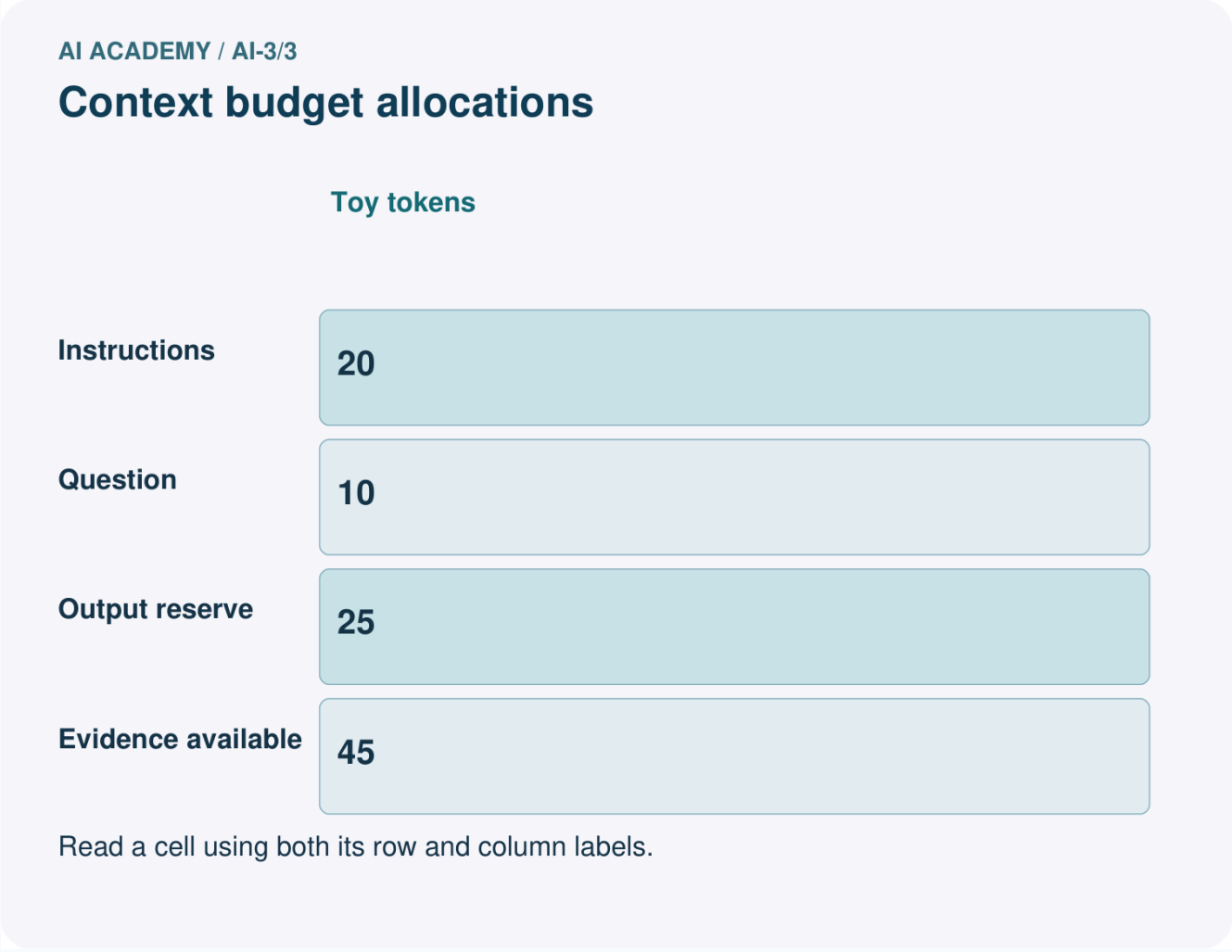
Pictorial case practice: Tokens and Context

> A fictional assistant has a small context budget for answering an event-capacity question. Explain the labelled step before checking the changed condition.

A fictional assistant has a small **context budget** for answering an event-capacity question.

Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-3/3

Tokens and Context

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Compute budget

100-20-10-25=45 evidence tokens.

2. Select relevant pair

A+B costs 38 and preserves the capacity exception.

3. Reject incomplete shortcut

A+C costs 30 but leaves out the exception that changes the answer.

4. Answer with scope

For this event, the declared cap is 12, with both A and B retained.

CHANGE ONE THING

Increase B's cost to 30 with all other costs fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Can A alone justify answering 20 seats after the exception is known?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Verified Study Buddy record.

Project: Verified Study Buddy

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional assistant has a small **context budget** for answering an event-capacity question.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? For this event, the declared cap is 12, with both A and B retained.

1. Answer with scope

2. Select relevant pair

3. Reject incomplete shortcut

4. Compute budget

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Answer with scope For this event, the declared cap is 12, with both A and B retained. A **context window** limits how much information is available in one model call.

Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement. Keep the relevant source relationship visible.

Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Increase B's cost to 30 with all other costs fixed. Which conclusion is supported by the supplied facts?

1. A+B costs 48 and no longer fits the 45-**token evidence** capacity.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

> Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Repair a missing building block

Review A Distribution of Possibilities (ai-3/2). Goal: Check whether a next-**token** table is a valid probability distribution. Short practice: Why is 0.625 not a broken probability sum?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
> | |
> |---|
> | |
> | |
> | |
> | |

WORKBOOK · 26 OF 26

Visual concept journey

> Explain each Tokens and Context picture from its labels, then record a relevant limit.

01. Tokens and Context: the case

AI ACADEMY / AI-3/3 · 01 · LOOK AT THE FACTS

Tokens and Context: the case

The situation

A fictional assistant has a small context budget for answering an event-capacity question.

Supplied evidence

Total budget 100 toy tokens; instructions 20, question 10, output reserve 25. Evidence capacity is the remainder.
Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

By the end, I can budget context and preserve evidence needed to interpret a requested fact.

Follow the labelled picture with the matching explanation. Open the vector diagram

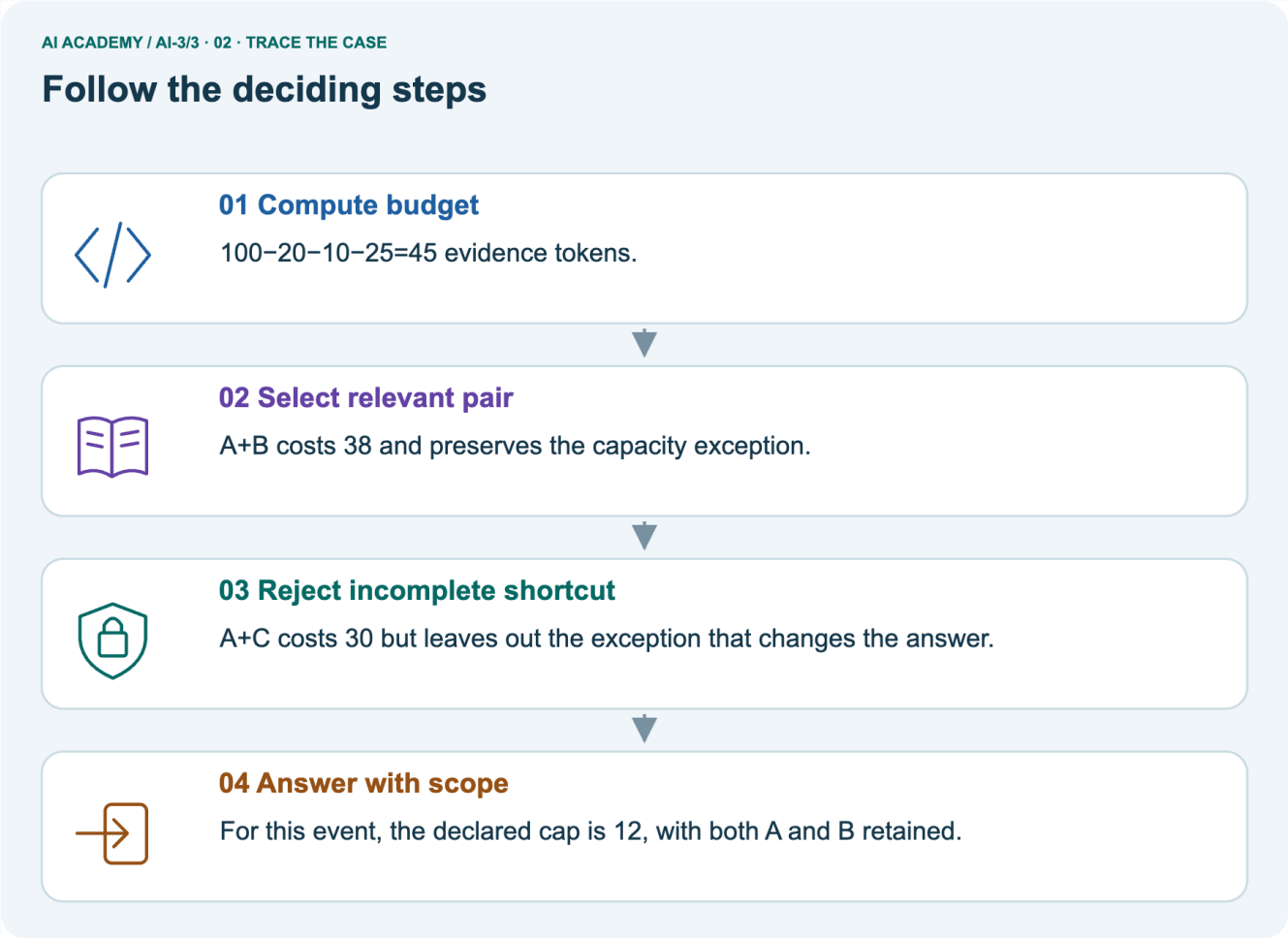
A fictional assistant has a small **context budget** for answering an event-capacity question. Supplied facts: Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute budget: 100−20−10−25=45 **evidence token**s. → Select relevant pair: A+B costs 38 and preserves the capacity exception. → Reject incomplete shortcut: A+C costs 30 but leaves out the exception that changes the answer. → Answer with scope: For this event, the declared cap is 12, with both A and B retained.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Context budget allocations



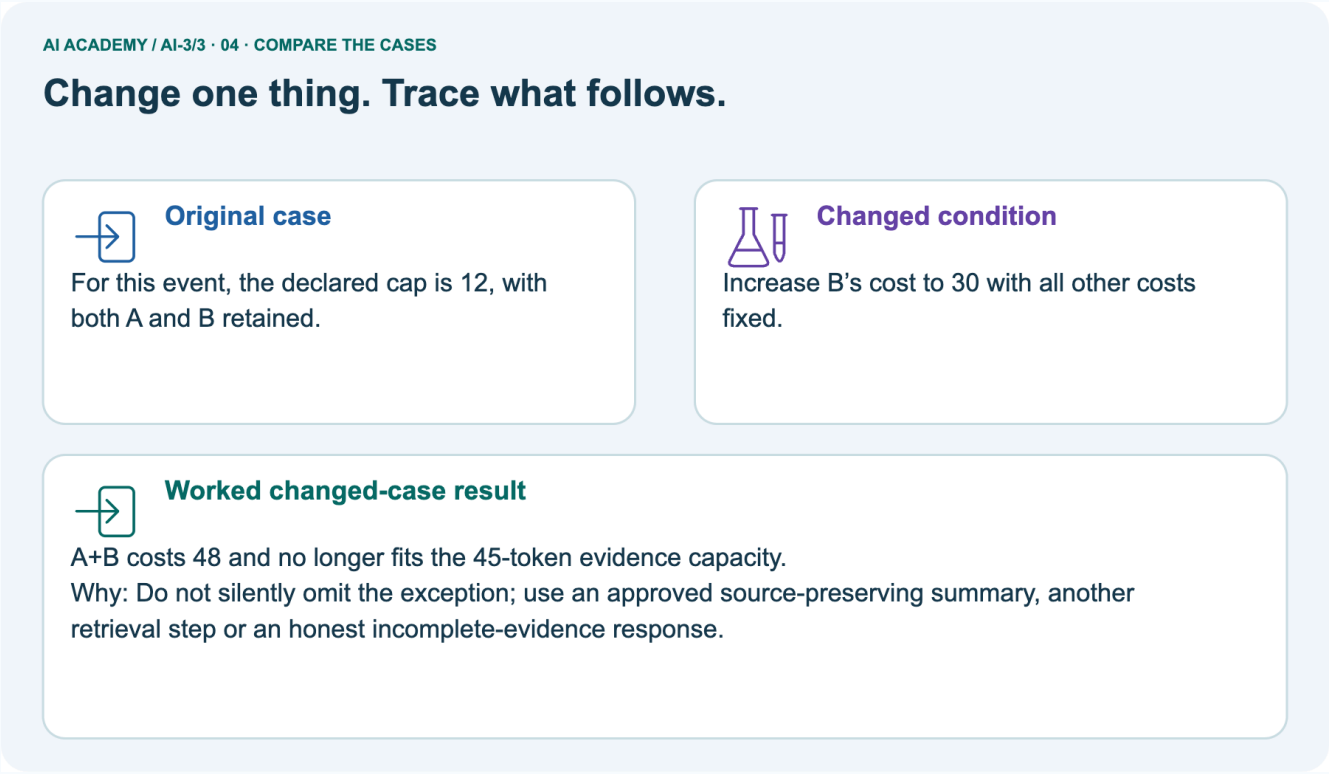
Follow the labelled picture with the matching explanation. Open the vector diagram

Context budget allocations. Instructions: Toy **token**s = 20. Question: Toy **token**s = 10. **Output** reserve: Toy **token**s = 25. **Evidence** available: Toy **token**s = 45.

Read the labels and **value**s. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: For this event, the declared cap is 12, with both A and B retained. Changed condition: Increase B's cost to 30 with all other costs fixed.

Expected result: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

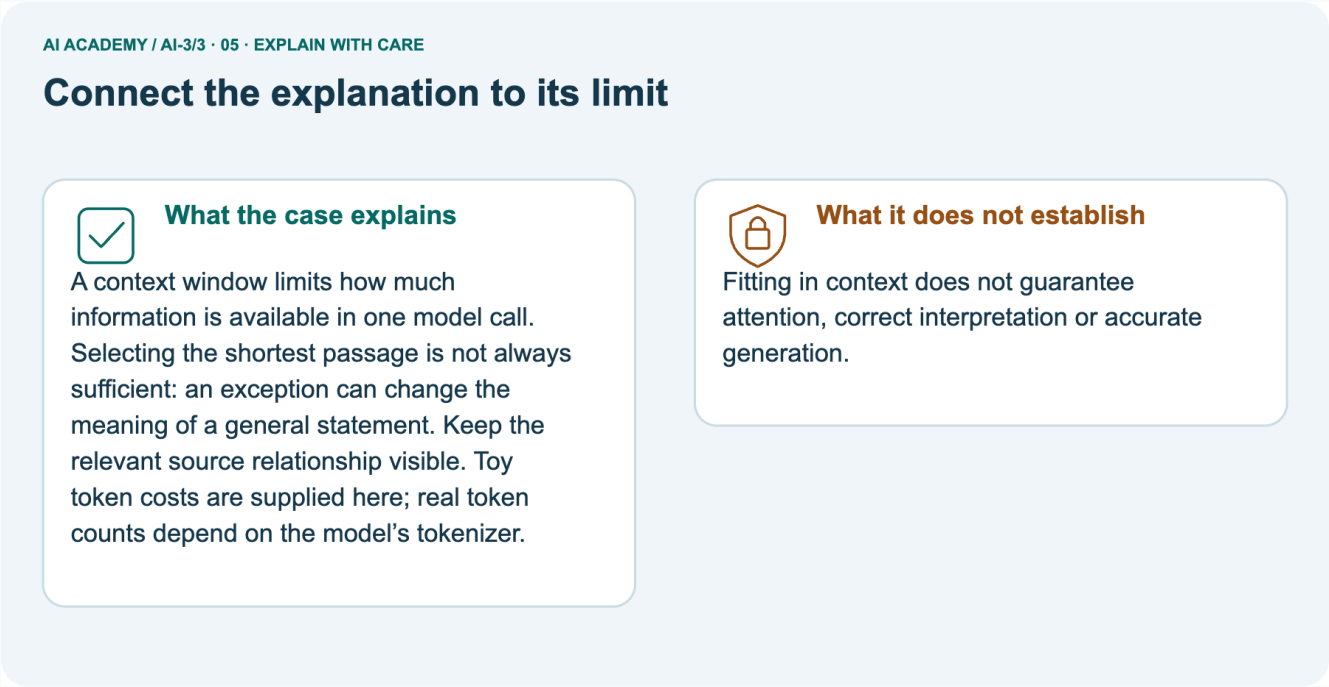
Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Predict before checking the worked changed-case result. Name what changed and what the

supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **context window** limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement.

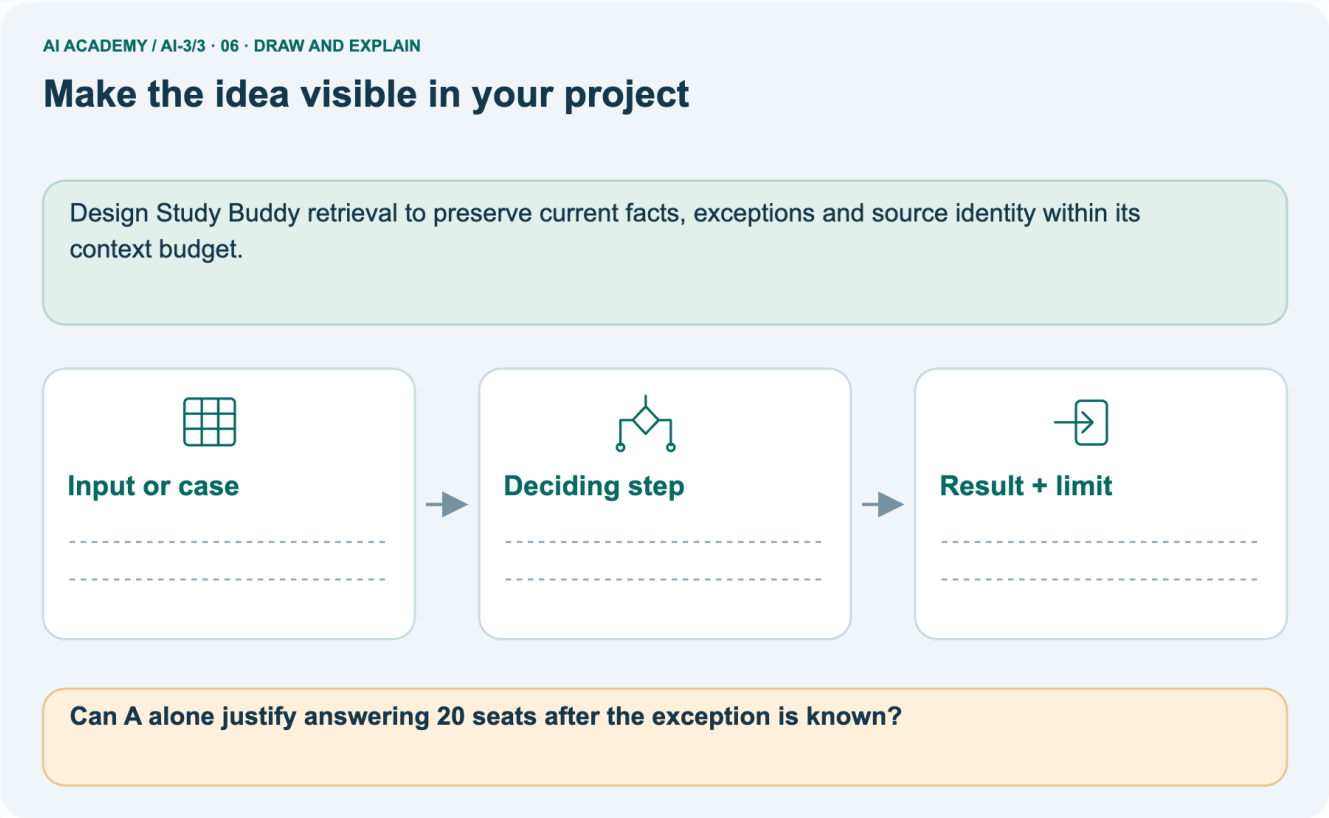
Keep the relevant source relationship visible. Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Limit: **Fitting** in context does not guarantee attention, correct interpretation or accurate generation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**. Draw your **input**, deciding step, result and limit.

Supported question: Can A alone justify answering 20 seats after the exception is known?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.